Problem statement and objectives

Context

UN Comtrade publishes trade *values*, but analysts first need to know **what data exist**: which reporter, which
year, which classification, how large, and when last released. That is the *Data Availability* export. It carries
no trade values, it is rarely analysed as a dataset, and its raw form is awkward to load: optional fields are
omitted rather than left blank, so each row is left-shifted (8-13 populated cells instead of 13).

Analytical questions

#	Question	Why it matters
RQ1	Can the export be reconstructed reliably into a tidy table?	Every later result depends on correct field assignment
RQ2	How did classification regimes, reporting volume and file size evolve, 1962-2025?	Tells analysts which edition covers which period and how data grew
RQ3	How timely are releases, and can timeliness be compared across eras when only the *last* release date exists?	Timeliness is a core data-quality dimension, but the field is biased by exposure
RQ4	Can file size be predicted from record count, and is that relationship stable over time?	Capacity planning and sanity-checking releases

Hypotheses tested

- H1: yearly record totals grow log-linearly (a constant compound rate).
- H2: bytes per record changed after 2010 (null: no change).
- H3: exposure-adjusted lag differs across HS editions (null: no difference).

Success metrics (bars chosen as reasonable thresholds, not pre-registered)

Metric	Target	Result
Rows reconstructed	100%	8,916 / 8,916
Agreement with an independent grammar	>= 99%	100% (0 violations); validator proven to catch the original bug
Random manual audit (60 rows, seed 0)	0 errors	0 errors (author-reviewed, not an independent annotator)
Categorical integrity	`typeCode` single-valued; notes only the upstream artefact	Pass
Growth quantified with an interval	CAGR + 95% CI	12.0% per year (10.9-13.1%), R^2 0.93
Per-record size test with effect size	p-value + effect size	median 13.0 -> 16.5 bytes, rank-biserial -0.54
Size model, temporal holdout (train <= 2010, test > 2010)	R^2 >= 0.95	0.973 (records), 0.982 (+ year)
Reproducibility	one command regenerates headline numbers; tests pass	`run_pipeline.py`, `run_stats.py`, 14 tests

Out of scope

Trade values, economic interpretation of record growth, and causal claims. The export has one source, so there is
no multi-table integration; joining external reference data (e.g. income groups) is listed under next steps.